LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose of test signals used in control systems, Step function, Impulse function, RAMP functions, Parabolic functions
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose of test signals used in control systems, Step function, Impulse function, RAMP functions, Parabolic functions.
5. Specific Objectives	20.1.1.1 Trainee will be able to classify control systems, feedback and its effect.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of signal generation on oscilloscope; Safety: High voltage precautions; Supervision: Instructor monitors equipment settings.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a test signal in a control system?
(b) How does feedback affect the stability of a control system?
(c) Why are standard test signals necessary for system analysis?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Purpose of test signals in control systems • Definition and characteristics of Step function • Definition and characteristics of Impulse function • Definition and characteristics of RAMP functions • Definition and characteristics of Parabolic functions	L & Ds	PPT, IB	15 min
(b)	• Comparison of signal magnitudes and durations • Application of Step function in system testing • Application of Impulse function in transient analysis • Application of RAMP functions in rate control • Application of Parabolic functions in acceleration	L & Ds	PPT, PDF	20 min
(c)	• Selection criteria for specific test signals • Impact of signal type on system response • Summary of signal classifications • Review of feedback effects on signal interpretation • Preparation for next lesson on unit step response	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a Step function and an Impulse function.
(b) Why is the RAMP function used for testing rate control systems?
(c) How does the Parabolic function relate to acceleration inputs?

12. Home Assignment and Follow Up	Review the classification of control systems and prepare a brief note on how feedback modifies the response to a Step function, linking concepts from the previous lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Unit step response of a control system, Transient response, Steady state response, Max overshoot, Rise time, Delay time and settling time.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min